

JOONWON LEE

QUANTITATIVE RESEARCH | STATISTICAL MACHINE LEARNING | MODEL RISK

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RESEARCH PROFILE

Statistics Ph.D. candidate developing scalable probabilistic models and diagnostics for high-dimensional dependent data.

Research combines sparse likelihoods, spectral methods, stochastic-process theory, and GPU/HPC implementation; industry experience spans financial risk monitoring and commercial insurance pricing.

EDUCATION

Rutgers University, Piscataway, NJ - Ph.D. in Statistics

Expected Dec 2026 | GPA: 3.8/4.0

University of Minnesota, Minneapolis, MN - M.S. in Statistics

Sep 2019 - Sep 2021 | GPA: 3.7/4.0

RESEARCH EXPERIENCE

Ph.D. Research - Scalable Inference and Diagnostics for High-Dimensional Space-Time Processes Sep 2023 - Present

Application domain: geostationary satellite total-column ozone, with 18,126 spatial cells and eight daytime snapshots per day. The statistical work is designed to transfer to any large, correlated panel or stochastic field where dense covariance methods are computationally infeasible.

- **Scalable likelihood:** Developed a nonseparable, advection-aware Vecchia approximation with 4x4 target blocks and a 6/4/3 temporal conditioning pattern. The fixed conditioning budget replaces dense $O(n^3)$ covariance operations with approximately linear scaling in n , enabling daily fits over 145,008 rows (about 141,000 observed after quality filtering).
- **Approximation and model selection:** Built max-min block-prefix experiments at 100, 200, 400, 600, 800, and all blocks to separate parameter instability caused by the Vecchia approximation from instability caused by covariance-family misspecification. Compared Matérn and generalized Cauchy models on July 2023-2025 data using likelihood paths and parameter convergence.
- **Conditional-eigen model-risk diagnostic:** Eigendecomposed small conditional covariance blocks, projected residuals into whitened eigen-coordinates, adjusted expected increments for fitted-mean leverage, and accumulated squared scores by conditional eigenvalue. A Brownian-bridge envelope turns the curve into a scale-resolved specification check rather than a single global fit statistic.
- **Controlled validation:** On a simulated 141,301-score day, the correctly specified model produced $D=0.86$ against a 95% cutoff of 1.358; wrong smoothness specifications produced $D=12.49$ and 13.45, and wrong nugget specifications produced $D=13.62$ and 21.35. Cross-checked behavior using full eigendecomposition on a max-min subset with 3,113 observed points.
- **Spectral distortion under missing data:** Derived how zero-imputation, structured observation masks, and resampling operators alter the expected periodogram and the pseudo-true variance, range, and nugget parameters of Debiased Whittle estimation. Developed a corrected-periodogram route for reducing the resulting scale and noise-floor bias.
- **Multivariate frequency diagnostic:** Constructed an eight-variate Fourier vector for hourly fields, computed the mask-aware finite-sample expected 8x8 cross-periodogram, applied Cholesky whitening, and profiled residual power by norm, latitude, longitude, and diagonal frequency. Validated the expected cross-periodogram over 1,000 simulations with MAD 0.00648.
- **Physics-aligned advection checks:** Used asymmetric one-hour cross-variograms and two-dimensional empirical ridge locations to test whether fitted advection vectors align with observed movement. Added near-minimum counts and relative-gap measures so flat or multimodal ridges are flagged as ambiguous rather than over-interpreted.
- **Research engineering:** Implemented reusable PyTorch likelihood engines, GLS mean profiling, L-BFGS optimization, CPU/GPU fallbacks, chunked conditional diagnostics, simulation generators with circulant embedding, and restartable HPC pipelines with cached fit artifacts.

SELECTED WORKING MANUSCRIPTS AND RESEARCH SOFTWARE

Spectral Distortion in Debiased Whittle Estimation under Structured Missingness. Working manuscript deriving random-mask, deterministic-mask, and resampling-operator effects on the expected periodogram; characterizes pseudo-true covariance parameters and a two-step bias-correction strategy.

Covariance of Empirical Periodogram Ordinates. Technical manuscript deriving exact Gaussian cross-frequency covariance under tapers and time-varying observation masks, with smooth, full-convolution, and structured-shrinkage estimators for finite-sample uncertainty.

Vecchia Conditional-Eigen Diagnostic Suite. Reusable PyTorch research software for blockwise whitening, leverage-adjusted score accumulation, Brownian-bridge specification bands, parameter-mismatch experiments, and full-eigen sanity checks.

Advection-Ridge and Cross-Variogram Diagnostics. Three-model comparison framework for July 2022-2025 that aligns empirical cross-variogram minima with fitted transport vectors and reports ridge ambiguity before interpreting direction or speed.

INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling Summer 2026

- Designed a sequential anomaly-detection framework for binary risk indicators by modeling cumulative evidence as a random walk and using Brownian-motion approximations to quantify stopping-time uncertainty.
- Structured simulation-based comparisons of fixed-sample and sequential rules around detection delay, false-positive/false-negative control, and the value of earlier decisions.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team Summer 2025

- Developed an end-to-end LightGBM pricing pipeline for mid-market property risk, including feature processing, GLM benchmarking, validation, and performance diagnostics over 2.46M+ records on AWS EC2.
- Used model diagnostics and feature attribution to translate predictive patterns into pricing and risk-segmentation recommendations for business stakeholders.

SELECTED QUANTITATIVE PROJECT

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants Apr 2025

- Built and evaluated a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven performance analysis.

TECHNICAL SKILLS

Programming and Systems: Python, NumPy, Pandas, PyTorch, SciPy, LightGBM, scikit-learn, R, SQL/MySQL, Linux, Git, AWS EC2, HPC/SLURM

Statistical Computing: MLE, Gaussian processes, Vecchia approximations, Matérn and generalized Cauchy covariance, spectral/Whittle methods, FFT and circulant embedding, GLS, L-BFGS, Monte Carlo simulation

Modeling and Validation: Stochastic processes, sequential testing, Brownian approximations, eigen diagnostics, cross-validation, GLM/GBM, model monitoring, missing-data diagnostics, directional semivariograms

COURSEWORK AND LANGUAGES

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)